

Model	F1 $\uparrow$	DAF $\uparrow$	Prec $\uparrow$	Acc $\uparrow$	MAE $\downarrow$	RMSE $\downarrow$	R ² $\uparrow$	Training Size	Model Type
CHGNet	0.94	5.29	0.88	0.88	0.06	0.11	0.81	1.6M (145.9K)	UIP-GNN
M3GNet	0.89	4.79	0.80	0.80	0.10	0.16	0.62	188.3K (62.8K)	UIP-GNN
MEGNet	0.64	2.84	0.47	0.47	0.33	0.36	-0.82	133.4K	GNN
CHGNet→MEGNet	0.64	2.83	0.47	0.47	0.36	0.38	-1.11	1.6M (145.9K)	UIP-GNN
M3GNet→MEGNet	0.64	2.82	0.47	0.47	0.36	0.38	-1.09	188.3K (62.8K)	UIP-GNN